Rajkumar Kishor Ramteke

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Profile

AI Engineer with about 2 years building agentic and generative AI products, and the Python and FastAPI services behind them. Work covers RAG precedent search on the Microsoft marketplace, a tool-calling legal chatbot, OCR annotation on briefs, and a 41-table lawyer platform. Lead developer on an ~INR 4 crore US-client sync of Qualtrics, Salesforce, and Tripleseat. Hands-on with Azure OpenAI, Azure AI Search, Document Intelligence, PostgreSQL, MongoDB, and prompt engineering.

Skills

AI: RAG, LLMs, agentic workflows, tool calling, Azure OpenAI, GPT-4, GPT-4o-mini, embeddings, reranking, grounding, prompt engineering, OCR
Backend: Python, FastAPI, REST, PostgreSQL, MongoDB, SQL, SQLAlchemy, Celery, Redis, JWT, RBAC, AsyncIO, Docker, GitHub Actions
Azure: App Service, Blob Storage, OpenAI, AI Search, Document Intelligence, resource migration
Integrations: Salesforce Apex, HMAC webhooks, OAuth 2.0, Microsoft SaaS marketplace, Razorpay, eCourts, Qualtrics, Tripleseat
Languages: Python, SQL, Apex, C++, C, JavaScript, Java

Experience

Software Engineer (AI/Backend) | AIOrdinate — Nov 2025 – Present

Remote

- **Lead developer, SF Giants, June 2026 – October 2026:** With one architect, automated Qualtrics, Salesforce, and Tripleseat. Booking and event updates had been typed by hand, and a prior vendor had not made the sync work. The automated path cut that handling time by about 90–95%. HMAC webhooks, OAuth 2.0, a claim queue, 2-minute dedup, and HTTP 429 retry. About an INR 4 crore engagement, the company's highest-value project.
  - Routed each request to Prospect, Waitlist, Tentative, or Definite with a 90-day lead window, a room-overlap check, and a 48-hour last-minute rule. Account, contact, owner, and manager ids map both ways, including Qualtrics inbound and the room-catalog API.
- **Knowledge transfer:** Wrote the admin docs for the booking path and onboarded junior engineers onto the Giants sync before moving to precedent search.
- **LexAIO agentic chatbot, April 2026 – Present:** A tool-calling agent. GPT-4o-mini reads the last 5 turns and routes to a law-dictionary hybrid search (top 5), an eCourts CNR lookup, or MasterDB, with 3 retries. GPT-4 answers only from that tool payload and writes follow-up questions. Language is detected on the query.
- **Chatbot APIs on both portals:** 13 client-portal routes for sessions, ask, case summary, and share-summary email. PostgreSQL stores detected language, confidence, follow-ups, and an English translation of each answer. Deployed the agent on Azure App Service with GitHub Actions.
- **Client portal:** Built the client-facing FastAPI service, 60+ JWT routes: signup and password reset, pinned cases, document folders (create, rename, move), task comments, calendar book and cancel, and Rocket.Chat. Court-order PDFs come from an eCourt scrape whose captcha is solved with Azure OpenAI vision.
- **Precedent finder, June 2026 – October 2026:** Built search over full judgments. A question, a saved matter, or an uploaded brief comes back as matching cases with IRAC, ratio decidendi, obiter, arguments for and against, and a 0–100 relevance score.
  - An orchestrator rewrites the query. Embedding search and rerank cover India, the US, the UK, and Australia and return the default 4 judgments from the top 20 chunks. Prompt work raised the confidence of the cases that are shown. A 40 confidence floor hides weak hits, and analysis retries under 0.30 retrieval or 0.35 grounding.
- **Microsoft marketplace:** Precedent search is sold as a Microsoft SaaS offer. The API does landing, activate, and lifecycle webhooks, then enforces plan entitlements and usage quotas. Direct checkout uses Razorpay. OCR covers 5 files at 10 MB. A 20-document cap stopped HTTP 503s and a shared Postgres pool stopped HTTP 500s.
- **Caseflow, lawyer platform:** Built the FastAPI and PostgreSQL API behind the lawyer product, with MongoDB for chat and search state. 41 tables and 140+ routes cover cases, clients, documents, meetings, and tasks, with JWT and role-based access. Migrated storage and app services onto Azure Blob, App Service, OpenAI, and AI Search. Celery workers and GitHub Actions deploys.
  - **Annotation and LODE:** Annotation, with Azure Document Intelligence, OCR, and Azure OpenAI, moved from about 75% to about 98% on bookmarks, the hierarchical index, and annexure references, then embeds them on download. LODE, with petitioner, respondent, and neutral context, reached about 96% extraction accuracy. A repeat still drops at an 85% similarity threshold.
- **Annotation page fix:** Handwritten page numbers on the brief did not match the PDF page, so a bookmark opened the wrong page. Preprocessing drops those handwritten numbers, and the prompt places the annotation on the page where the heading is found.

Technologies: Python, FastAPI, PostgreSQL, Celery, Redis, Azure OpenAI, Azure AI Search, Salesforce Apex, Docker, GitHub Actions.

Software Engineer Trainee | Cisco Systems India Pvt Ltd — Nov 2024 – Nov 2025

Bangalore

- Built FastAPI create, read, and update endpoints and PostgreSQL schemas for an internal ops dashboard of device records, configs, and status, with request validation and error handling.
- Wrote the SQL for list and filter screens, added JWT login, and debugged failing endpoints and integration tests in Git reviews with senior engineers.

Technologies: Python, FastAPI, PostgreSQL, SQL, REST, JWT, Git.

Projects

DeepDDI – Drug–Drug Interaction Prediction | IIIT Bhubaneswar, 2024

- Implemented three DrugBank interaction models. A SMILES Conv1D baseline reached 71.3% accuracy and 0.709 AUC. RDKit molecular graphs, 79 atom features and 10 bond features, lifted graph convolution to 81.0% accuracy and 0.804 AUC.
- Added Laplacian positional encoding and graph attention. Accuracy moved from 71.3% to 86.2%, and AUC from 0.709 to 0.850, the strongest of the three models.

Technologies: Python, PyTorch, RDKit, graph attention, Conv1D, DrugBank.

Education

IIIT Bhubaneswar — Dec 2020 – June 2024

B.Tech, Computer Engineering | CGPA 7.44 / 10 | DSA, OS, DBMS, OOP, Machine Learning

Achievements

- LeetCode Knight, 500+ DSA problems. CCNA. Coursera Supervised Machine Learning.